

AI-Powered Secure Data Analyzer

Intelligent Analytics Platform with Built-in AI & Cybersecurity

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1. Problem Statement

Organizations generate massive data but face two key issues: extracting insights requires expertise, and understanding data risk is difficult. BI tools like Power BI, Tableau provide visualization and access control, but do not analyze datasets for security risks. With breaches averaging **USD 4.88M** (IBM, 2024), there is a need for a platform that combines analytics with AI-driven risk detection.

2. Dataset Strategy

Model	Training Data
Anomaly Detection (Isolation Forest + LSTM)	KDD Cup / NSL-KDD datasets
Data Quality Classifier (distilBERT)	Synthetic dirty vs clean datasets
NL Query Model (T5-small)	WikiSQL + Spider datasets
Risk Score Regressor (XGBoost)	Auto-generated dataset from uploaded files

The risk dataset is generated dynamically using: PII density (Presidio), anomaly scores, and missing value ratios.

3. Proposed Solution

Phase 1 — Core System

The system allows users to upload CSV/Excel files and automatically performs:

- **Data Profiling:** Automatic column analysis using ydata-profiling.
- **AI Data Cleaning:** Fine-tuned distilBERT detects dirty data; users approve corrections via Human-in-the-Loop interface.
- **Smart Dashboard:** Auto-generated charts using Plotly.js.
- **Natural Language Query:** T5 model converts user queries into SQL/pandas operations.
- **Forecasting:** Prophet-based time-series predictions.
- **Security Engine:** Anomaly detection using Isolation Forest and LSTM.
- **Risk Scoring:** XGBoost generates a 0–100 risk score per dataset.

Phase 2 — Security & Enhancement

- **MFA + RBAC:** Role-based access
- **AES-256 Encryption:** Secure storage and communication.
- **Prompt Injection Protection:** Secure LLM interac-

tion.

- **Model Improvement:** Continuous retraining using new data.
- **Audit Logs:** Transparent AI decision tracking.

4. Deliverables

Phase 1:

- Full-stack web application
- 4 trained AI models with evaluation reports
- Interactive dashboard with 12+ visualizations
- Natural language query system
- Data cleaning pipeline
- Forecasting module
- Risk scoring engine

Phase 2:

- Security implementation (MFA, RBAC, Encryption)
- Model retraining pipeline
- Prompt injection protection
- PDF report generation
- Complete system integration

5. Tools & Technologies

Layer	Technology
Frontend	React 18, Plotly.js, Monaco Editor
Backend	FastAPI, Python 3.11
AI Models	scikit-learn, PyTorch, Transformers, XGBoost
LLM API	Google Gemini
Data Tools	pandas, numpy, Prophet
Security	AES-256, Presidio
Database	Firebase Auth, Firestore
Deployment	Vercel, Render

7. Innovation & Differentiation

This project uniquely combines:

- Custom-trained AI models
- Natural language analytics
- Real-time anomaly detection
- Built-in cybersecurity features

Unlike traditional BI tools, it integrates **analytics + AI + security** into one unified platform, making it accessible, intelligent, and secure.